# Birla Institute of Technology and Science Pilani

K.K. Birla Goa Campus

**AY 2025–26, Semester II**

**Course Handout (Part-II) for *Combinatorial Optimization* (CS F431)**

**Course:** ***Combinatorial Optimization*** **(CS F431)**
**Instructor-in-charge:** ***Dr. Diptendu Chatterjee*** (diptenduc@goa.bits-pilani.ac.in) **(Chamber : D264)**

This document is **Part II** of the handout for **Combinatorial Optimization (CS F431)**. You can find **Part I** of the handout appended to the timetable for the current semester. Part-I contains the general operational details for all the courses.

**Prerequisite**: Preferably Data Structures and Algorithms / Foundations of Data Structures and Algorithms.

**Course Objectives:**

- Learn about the identification, formulation, and mathematical foundation of combinatorial optimization.
- Analyzing the hardness of different combinatorial optimization problems and categorizing them.
- learn various algorithmic approaches to solve or approximate some of the combinatorial optimization problems.

**Text Book(s)/ Reference Book(s):**

**B1 Combinatorial Optimization: Algorithms and Complexity** by *Christos H. Papadimitriou* and *Kenneth Steiglitz*, Publisher: Prentice-Hall of India Pvt. Ltd. 2003.

**R1 Combinatorial Optimization: Theory and Algorithms** by *Bernhard Korte* and *Jens Vygen*, (6th Editon) Publisher: Springer.

**Makeup Policy:**

- Permission of the Instructor-in-Charge is required to take a make-up.
- Make-up applications must be personally given to the instructor in charge.
- A make-up test shall be granted only in genuine cases where, in the Instructor's judgment, the student would be virtually or physically unable to appear for the test.
- In case of an unanticipated illness preventing a student from appearing for a test, the student must present a Medical Certificate from the medical center.

Requests for make-up for the comprehensive examination under any circumstances can only be made to In-charge, the Instruction Division.

**Course Plan:**

| No. of Lectures | Topics | Reference |
|---|---|---|
| | **Module 1: Introduction** | |
| 2 | Optimization Problems | **B1** Chapter 1 |
| 2 | Combinatorics (Basics) | Lecture Notes |
| 2 | Graphs (Basics) | Lecture Notes and **R1** Chapter 2 |
| 2 | Linear Algebra (Basics) | Lecture Notes |
| | **Module 2: LP and ILP** | |
| 4 | Formation of LP, Feasibility, Linear and Affine space | **B1** Chapter 2 and **R1** Chapter 3 |
| 1 | Duality | **B1** Chapter 3 and **R1** Chapter 3 |
| 2 | ILP and Hardness | **B1** Chapter 13 and **R1** Chapter 5 |
| 3 | **Module 3: Path Optimization** | **R1** Chapter 7 |
| 3 | **Module 4: Flow Optimization** | **R1** Chapter 8 |
| 3 | **Module 5: Matching Problems** | **B1** Chapter 10-11, **R1** Chapter 10-11 |
| 3 | **Module 6: NP-Completeness** | **B1** Chapter 15-16 |
| 3 | **Module 7: Approximation Algorithms** | **B1** Chapter 17 and **R1** Chapter 16 |
| 3 | **Module 8: Bin Packing Problems** | **R1** Chapter 17-18 |
| 3 | **Module 9: Travelling Salesman Problem** | **R1** Chapter 21 |
| 3 | **Module 10: Matroids** | **R1** Chapter 13 |

**Evaluation Scheme:**

| **Component** | **Percentage** | **Date** |
|---|---|---|
| Quizzes | 40% | TBA |
| Mid-semester Exam | 30% | 10.3.26 (4 PM – 5:30 PM) |
| Comprehensive Exam | 30% | 7.5.25 (AN) |

**Chamber Consultation Hours:** To be announced in the class.